FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XII

JEE (Main)-2021

TEST DATE: 20-06-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

Sol. Newton's law of cooling.
Rate of decrease of temperature $\propto$ temperature difference between object and surrounding.

2. A

Sol. By equation of equilibrium,

$2T \sin 45° + Mg = \frac{4}{3}\pi R^3 \rho_w g$

3. C

4. C

Sol. $\Delta V = \left|\frac{d\phi}{dt}\right| = (\text{Area}) \times \frac{dB}{dt} = \frac{\pi R}{2}\left(\frac{R}{2}\right)(B_o)$

$\Delta V_{AB} = \frac{\pi R^2}{4} B_o$

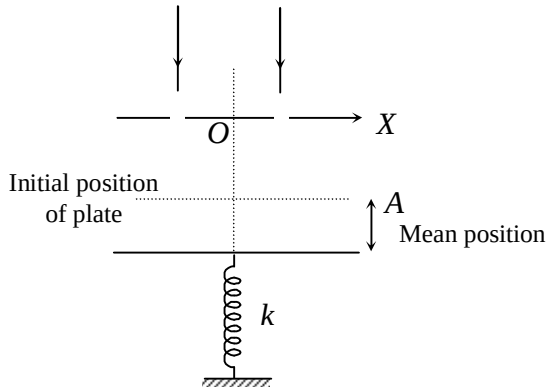
5. A

Sol. $\beta = \dfrac{(D + y)\lambda}{d}$

$\dfrac{d\beta}{dt} = \dfrac{\lambda}{d}\dfrac{dy}{dt} = \dfrac{\lambda}{d}v$

At mean position $v = A\omega = \dfrac{mg}{k}\sqrt{\dfrac{k}{m}} = g\sqrt{\dfrac{m}{k}}$

$\dfrac{d\beta}{dt} = \dfrac{\lambda g}{d}\sqrt{\dfrac{m}{k}}$



6. A

Sol. $\dfrac{1}{f} = (\mu - 1)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)$

On solving,

$\dfrac{1}{f_1} = \dfrac{-1}{40}, \dfrac{1}{f_2} = \dfrac{1}{60}$

For combination & silvering,

$-\dfrac{1}{F} = \dfrac{2}{f_1} + \dfrac{2}{f_2} - \dfrac{1}{f_m}$

F = 60 cm

$\dfrac{1}{v} + \dfrac{1}{u} = \dfrac{1}{F}$

On solving with u = –40 cm, F = 60 cm we get V = 24 cm

$v_Z = -\dfrac{v^2}{u^2}V_o = 1.8\,\hat{i}$

7. C

Sol. $\dfrac{1}{2}2k(2R\theta)^2 + \dfrac{1}{2}I\omega^2 = \text{constant}$

$4kR^2\theta^2 + \dfrac{7}{10}MR^2\omega^2 = \text{constant}$

$4kR^2 2\theta\dfrac{d\theta}{dt} + \dfrac{7}{10}MR^2 2\omega\dfrac{d\omega}{dt} = 0 \quad \Rightarrow \quad T = \pi\sqrt{\dfrac{7M}{10k}}$

8. A

Sol. By Bernoulli's theorem: $P + \dfrac{1}{2}\rho v^2 = \dfrac{P}{2} + \dfrac{1}{2}\rho v_1^2$

$\Rightarrow \quad v_1 = \sqrt{v^2 + \dfrac{p}{\rho}}$

9. B

Sol. Apply Newton's IInd law in y direction and use,

$$a_y = \frac{v_y dv_y}{dy}$$

10. C

Sol. $T = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$; $g_{eff} = g + \frac{qE}{m}$ (with electric field)

11. C

Sol. Case (1) Assume that mass m is accelerating upward.

$$\therefore \quad T - mg = \frac{mg}{10} \qquad \Rightarrow \qquad T = \frac{11mg}{10}$$

$$\therefore \quad nmg - T = \frac{nmg}{10} \qquad \Rightarrow \qquad nmg - \frac{11mg}{10} = \frac{nmg}{10}$$

$$\frac{9}{10}n = \frac{11}{10} \text{ and } n = \frac{11}{9}$$

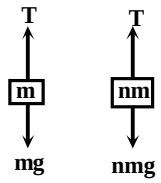


Figure 1

Case (2) Assume

If m mass is moving downward

$$mg - T = \frac{mg}{10} \Rightarrow \qquad T = \frac{9mg}{10}$$

$\therefore$ T – nmg = nmg/10

$$\Rightarrow \quad \frac{9}{10}mg - nmg = \frac{nmg}{10}$$

$$\frac{9}{10} = \frac{n}{10} + n = \frac{11n}{10}$$

$\therefore$ n = 9/11

12. B

Sol. $\ell = \ell_1 + \ell_2 \pm 2\Delta\ell = 3.80 \pm 0.02$

13. C

Sol. $E = \frac{1}{2}mv^2$

$$\Rightarrow [m] = E^1 v^{-2} T^0$$

14. A

Sol. In x direction

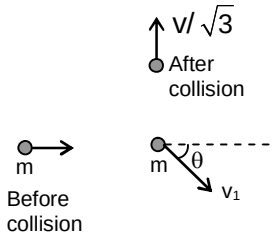
$mv = mv_1\cos\theta$ …(1)

In y direction

$0 = m\frac{v}{\sqrt{3}} - mv_1\sin\theta$ …(2)

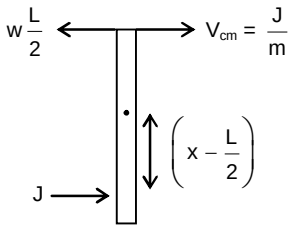
Square and add

$V_1 = \frac{2v}{\sqrt{3}}$



15. C

Sol. $J\left(x-\frac{L}{2}\right)=I\omega$



$\omega\frac{L}{2}=V_{cm}=\frac{J}{M}$; $\frac{J\left(x-\frac{L}{2}\right)\frac{L}{2}}{I}=\frac{J}{m}$

$I=\frac{ML^2}{12}$

16. A

Sol. Acceleration will be least in part ACB as its distance from the centre of earth is maximum and acceleration will be maximum in part ADB as its distance from centre of earth is minimum.

17. C

Sol. $A_r=\frac{|v_1-v_2|}{v_1+v_2}\times A_i$; $v_1=\sqrt{\frac{T}{\mu}}$; $v_2=\sqrt{\frac{T}{4\mu}}$; $v=f\lambda$

18. A

Sol. $4\pi R^2=6l^2$ and $ms\left(\frac{-dT}{dt}\right)=\sigma\varepsilon AT^4 \Rightarrow$ Rate of cooling (R) $\propto \frac{1}{m}$

$\Rightarrow \frac{R_1}{R_2}=\frac{\rho(l)^3}{\rho\left(\frac{4\pi}{3}R^3\right)}=\frac{3}{4\pi}\left(\frac{l}{R}\right)^3=\left(\frac{3}{4\pi}\right)\left(\frac{4\pi}{6}\right)^{3/2}=\sqrt{\frac{\pi}{6}}$

19. A

Sol. $y_1=a\cos\omega_1 t$ and $y_2=a\cos(\pi+\omega_2 t)$

For same phase $\omega_1 t=\pi+\omega_2 t$

$t=\frac{\pi}{\omega_1-\omega_2}=\frac{\pi}{\frac{2\pi}{3}-\frac{2\pi}{7}}=\frac{21}{8}s$

20. C

Sol. As $\gamma > 1$

for $TV^{\gamma-1}$ = constant $\frac{dT}{dV}<0$ $\frac{d^2T}{dV^2}>0$

and for $T^{\gamma}=KP^{\gamma-1}$

$\frac{dT}{dP}>0$ $\frac{d^2T}{dP^2}<0$

SECTION – B

21. 00000.35

Sol. By momentum conservation $(0.98+0.02)v = 0.02\times 3$

v = 0.06 m/sec

$\frac{1}{2}mv^2 = \frac{1}{2}KA^2$

$A = \sqrt{\frac{mv^2}{K}} = \sqrt{\frac{1\times 0.06^2}{300}} = 0.0035$ m = 0.35 cm

22. 00000.50

Sol. Writing the general expression for y in terms of x as

$y = \frac{1}{1+(x-vt)^2}$ at t = 0, $y = 1/(1+x)^2$.

At t = 2 s, $y = \frac{1}{1+[x-v(2)]^2}$

Comparing with the given equation we get 2v = 1 and v = 0.5 m/s.

23. 00000.25

Sol. $\Delta V = h\times A$

$10^{-6}\left(18\times 10^{-5}-3\times 10^{-5}\right) = 3\times 10^{-3}\pi\left(\frac{d}{2}\right)^2$

$d = 2.52\times 10^{-4}$m

24. 00002.40

Sol. V_x = 18 cos 60° = 9 m/s.

$V_{1y} = \sqrt{15^2-9^2} = 12$ m/s V_{2y} = –12 m/s

$\therefore\ \Delta t = \frac{-12-(12)}{-10} = \frac{12}{5}$sec.

25. 00001.60

Sol. Block will return after maximum elongation.

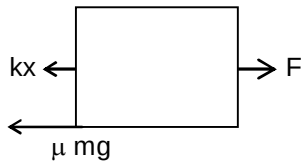
i.e. $F.x_{max} - ½\ Kx^2_{max} - \mu\ mgx_{max} = 0$

$x_{max} = \frac{2(F-\mu mg)}{k} = \frac{8\mu mg}{k}$

So, block will finally come to rest while returning i.e. $v = 0$ & $a = 0$

By work energy theorem while returning

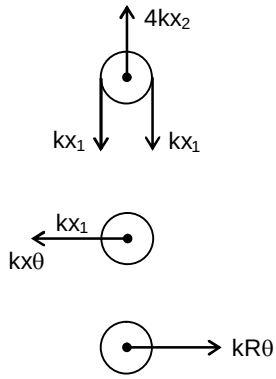
$-\left(\frac{1}{2}kx^2 - \frac{1}{2}kx^2_{max}\right) - (F+\mu mg)(x_{max}-x) = 0 \Rightarrow, x = \frac{4\mu mg}{k}$



When the block comes to rest for the second time it will have a tendency to move toward right. But friction will act towards left and hence stopping the block instantly.

26. 00008.56

Sol. When the cylinder is displaced towards right, (θ is angle of rotation of diameter of the cylinder about the contact point) then



$x_1 + 2x_2 = R\theta$

$x_1 + x_1 = R\theta$

$x_1 = \frac{R\theta}{2}$

$\tau = -[kx_1R + kR\theta R]$

$\frac{3}{2}mR^2\alpha = -\left[\frac{KR^2\theta}{2} + KR^2\theta\right]$

$\alpha = -\frac{k}{m}\theta \quad ; \quad t_1 = \pi\sqrt{\frac{m}{k}}$

When cylinder moves towards left, then

Torque about the contact point

$-kR^2\theta = \frac{3}{2}mR^2\alpha$

$\alpha = -\left(\frac{2}{3}\frac{k}{m}\right)\theta \quad ; \quad t_2 = \pi\sqrt{\frac{3m}{2k}}$

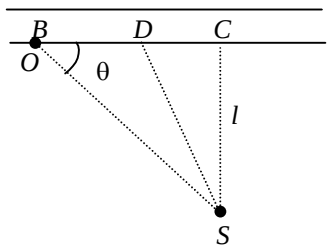
So, time period of oscillation

$T = t_1 + t_2$

$= \frac{\pi}{2}\left(3 + \sqrt{6}\right) = 5.56$ sec.

27. 00136.00

Sol. Let the bus be at O when it sends a signal that is detected by the detector as of frequency = 1500 Hz



$\therefore \quad f' = \left(\frac{v}{v - v_B\cos\theta}\right)\times 1000 = 1500$

$\therefore \quad \cos\theta = \frac{\sqrt{3}}{2} \quad \Rightarrow \quad \theta = 30°$

By the time signal reaches at S the bus reaches at D. Let this time be t_0

$\therefore \quad t_0 = \frac{OS}{v} = \frac{l\,\text{cosec}\,\theta}{v} \quad \ldots (1)$

Now man fires and the bullet reaches C in time t_1 (say). In the same time bus moves from D to C

$\therefore \quad t_1 = \frac{l}{u}$; where u = speed of bullet

Also, OD + DC = $l\cot\theta$

$v_B t_0 + v_B t_1 = l\cot\theta$

$v_B\left(\frac{l\,\text{cosec}\,\theta}{v}\right) + v_B(l/u) = l\cot\theta$

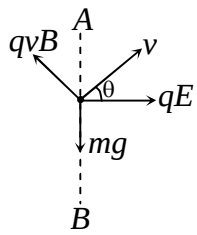
$\frac{2}{3\sqrt{3}}\times 2 + \frac{2}{3\sqrt{3}}\frac{v}{u} = \sqrt{3}$

$\therefore \quad \frac{v}{u} = \frac{5}{2}$

$\therefore \quad u = \frac{2}{5}v = \frac{2}{5} \times 340 = 00136.00$ m/s

28. 00019.60

Sol. When crosses *AB*



$qvB_0 \cos\theta = mg$

$qvB_0 \sin\theta = qE$

$\tan\theta = \frac{qE}{mg} = 1$

$\theta = \frac{\pi}{4}$

along horizontal $v\cos\theta = \frac{qE}{m}t_0$

$u - gt_0 = v\sin\theta$

$u = \left(g + \frac{qE}{m}\right)t_0 = 2gt_0$

29. 00000.22

Sol. Minimum separation = length of perpendicular dropped on $\vec{V}_{12}$.

30. 00028.20

Sol. Binding energy = (Δm) × 931.5 MeV

= [2 × 1.00785 + 2 × 1.00866 – 4.00274] × 931.5

= 28.2 MeV

Chemistry

PART – B

SECTION – A

31. C
Sol. One mole of NH_3 occupy 22.4 L at NTP, not at 300 K and 1 atm.

32. D
Sol. Sulphur forms two sigma bonds and with two lone pairs.

33. B
Sol. The unit of k will be $mol^{-1/2}L^{1/2}s^{-1}$.

34. D
Sol. K_a increases only by increasing temperature.

35. B
Sol. One Na^+ is removed and remaining three (effective number per unit cell) are present.

36. C
Sol. The fraction is $\frac{2}{3}$

37. A
Sol. Aliphatic amines are stronger base than aromatic amines

38. B
Sol. The configuration of carbon atoms at 3, 4, 5 positions should be same.

39. B
Sol. Molecular mass: $CaCl_2 = 111$, $AlCl_3 = 133.5$
$NaCl = 58.5$, $KCl = 74.5$
$\Delta T_b = K_b im$
$\therefore \Delta T_b \alpha i \times m$
$\therefore$ values of $i \times m$

$$CaCl_2 = \frac{w}{111} \times 3 = \frac{w}{37}, AlCl_3 = \frac{w}{133.5} \times 4 = \frac{w}{33.37}$$

$$NaCl = \frac{w}{58.5} \times 2 = \frac{w}{29.25}, KCl = \frac{w}{74.5} \times 2 = \frac{w}{37.25}$$

$\therefore$ The required order: $NaCl > AlCl_3 > CaCl_2 > KCl$

40. C
Sol. Beta keto acids undergo decarboxylation on heating.

41. B
Sol. P = CaO, Q = Ca_3N_2, R = $Ca(OH)_2$, S = NH_3, T = $CaCO_3$, U = NH_2CONH_2

42. B
Sol. $$Ni(\text{impure}) + 4CO \longrightarrow Ni(CO_4) \xrightarrow{\Delta} Ni(\text{pure}) + 4CO$$

43. B

Sol. Na_2CO_3 decomposes above 1100^oC

44. B

Sol. $Cl_2 + FeCl_3 \longrightarrow FeCl_4^- + Cl^+$

The electrophile Cl^+ substitutes a hydrogen atom from benzene.

45. C

Sol. Phosphates accelerate the growth of plants in water, which consume the dissolved oxygen

46. D

Sol. At normal boiling point (500 K), the following equilibrium exists between the two physical states at 1 atm.

$X(\text{liquid}) \rightleftharpoons X(\text{gas})$

For the equilibrium $\Delta G = 0$

If the pressure increases the change shifts to backward direction for which ΔG = +ve.

If pressure decreases the change shifts towards forward direction, for which ΔG = -ve.

If temperature increase, the change will proceed towards forward direction, for which ΔG = -ve.

47. C

Sol. Increases in pressure or addition of C affect the second equilibrium hence concentration of A_2 changes which further affect the first equilibrium. Addition of inert gas at constant volume does not affect the equilibrium

48. B

Sol. $AlF_3 + 3KF \longrightarrow K_3[AlF_6]$

For this reaction, maximum concentration of F^- ions is obtained from KF.

49. B

Sol. Q = It = (965 $\times$ 100) cou = 1 F

1 F deposits (1 g eq of Ag = 108 g)

50. C

Sol. Only size decreases due to lanthanide contraction.

SECTION – B

51. 00005.00

Sol. $$\frac{\lambda_{(Be^{4+})}}{\lambda_{(He^{2+})}} = \frac{\frac{h}{m(Be^{4+})U}}{\frac{h}{m(He^{2+})U}} = \frac{m(He^{2+})}{m(Be^{4+})} = \frac{4}{9} = \frac{x}{y}$$

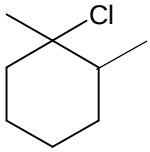
$\therefore$ y – x = 5

52. 00041.57

Sol. $\Delta G^o = -RT\ell nK_P$

= [-(8.314)(1000)(-5)] Joule = 41.57 kJ

53. 00146.50

Sol. The alkyl chloride is 

54. 00050.50

Sol. $CH_3Cl + KOH \longrightarrow CH_3OH + KCl$

Molar mass of CH_3Cl = 50.5 g mol^{-1}

55. 00149.50

Sol. CH_3SiCl_3 forms cross-linked silicones.

56. 06235.51

Sol. $K.E = \frac{3}{2}nRT = \frac{3}{2}(1)(8.314)(500) = 6235.51\ J$

57. 00416.00

Sol. $2T\ell OH \xrightarrow{\Delta} T\ell_2O + H_2O$

Molar mass of $T\ell_2O$ = 416 g mol^{-1}.

58. 00119.50

Sol. $CHCl_3 + NaOH \xrightarrow{HCl} HCOOH + H_2O + NaCl$

Molar mass of $CHCl_3$ = 119.5

59. 00000.80

Sol. Rate = $\frac{1}{2}\frac{d[A]}{dt}$

or 0.4 = $\frac{1}{2}\frac{d[A]}{dt}$

$\therefore \frac{d[A]}{dt} = 2 \times 0.4 = 0.8$

60. 00005.70

Sol. $pH = p^{K_a} + \log\frac{[CH_3COONa]}{[CH_3COOH]} = 4.7 + \log\frac{10}{1} = 4.7 + 1 = 5.7$

Mathematics

PART – C

SECTION – A

61. D

Sol. Equation of chord of the circle $x^2 + y^2 = 25$, whose midpoint is P (h, k)

$T = S_1$

$xh + yk - 25 = h^2 + k^2 - 25$

$\Rightarrow xh + yk - h^2 - k^2 = 0$ ……(1)

This chord is tangent to the circle

$x^2 + y^2 - 20x - 20y + 175 = 0$

Length of perpendicular from centre = Radius

$$\left|\frac{10h + 10k - h^2 - k^2}{\sqrt{h^2 + k^2}}\right| = 5$$

$$\Rightarrow \left(10h + 10k - h^2 - k^2\right)^2 = 25\left(h^2 + k^2\right)$$

$$\left(10x + 10y - x^2 - y^2\right) = 25\left(x^2 + y^2\right)$$

62. A

Sol. Clearly two sides are $x - 2y + 1 = 0$ and $x + y = 0$

As orthocenter lies on $x - 2y + 1 = 0$ is right angled (let at B)

Third side passes through (1, 1) and is perpendicular to $x - 2y + 1 = 0$

$\Rightarrow$ Third side is $2x + y - 3 = 0$

$$\therefore \text{ Circum – radius } = \frac{1}{2}AC = \frac{5\sqrt{2}}{3}$$

63. C

Sol. $x_1 + x_2 = 17$

$2t_1^2 + 2t_2^2 = 17$

$2\left(t_1^2 + t_2^2\right) = 17$

$$\left(t_1 + t_2\right)^2 - 2t_1t_2 = \frac{17}{2}$$

$x_1x_2 = 11$

$at_1^2at_2^2 = 11\left(a = 2\right)$

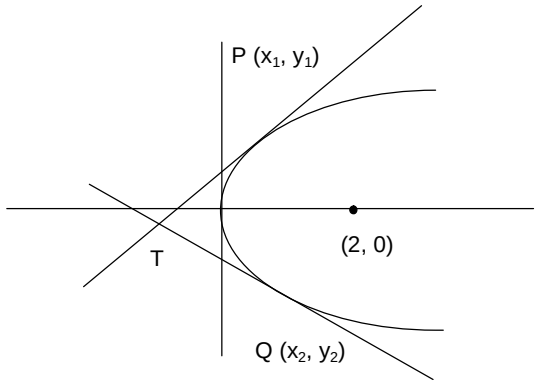
$4t_1^2t_2^2 = 11$

$$t_1^2t_2^2 = \frac{11}{4}$$

$T = \left[at_1t_2, a\left(t_1 + t_2\right)\right]$

$T = \left[2t_1t_2,\ 2\left(t_1 + t_2\right)\right]$

S : focus (2, 0)



$$TS = \sqrt{(2t_1t_2 - 2)^2 + 4(t_1 + t_2)^2}$$
$$= \sqrt{4(t_1t_2 - 1)^2 + 4(t_1 + t_2)^2}$$
$$= 2\sqrt{(t_1 + t_2)^2 + (t_1t_2)^2 + 1 - 2t_1t_2}$$
$$= 2\sqrt{\frac{17}{2} + 1 + \frac{11}{4}} = 2\sqrt{\frac{34 + 4 + 11}{4}} = \sqrt{49} = 7$$

64. B

Sol. $f(x) = \begin{cases} \cos x & : x > 1 \\ \cos x & : x < 1 \\ k & : x = 1 \end{cases}$

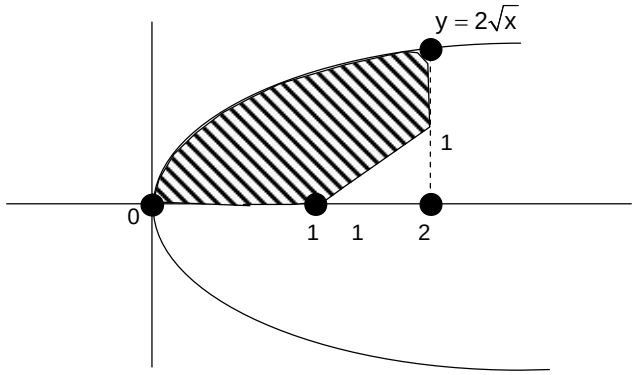
65. C

Sol. $y = [x](x - 1)$

$$= \begin{cases} 0 & 0 \le x < 1 \\ x - 1 & 1 \le x < 2 \end{cases}$$

$$\text{Area} = \int_0^2 2\sqrt{x}\,.dx - \frac{1}{2}(1)(1)$$

$$= \left(\frac{4x^{3/2}}{3}\right) - \frac{1}{2} = \frac{8\sqrt{2}}{3} - \frac{1}{2}$$



66. C

Sol. $$\int \frac{\sin^2 x \cos^2 x}{\left(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x\right)^2} dx$$

$$\int \frac{\sin^2 x \cos^2 x}{\left(\sin^3 x + \cos^3 x\right)^2} dx$$

$$\int \frac{\tan^2 x . \sec^2 x}{\left(1 + \tan^3 x\right)^2} dx$$

Put $1 + \tan^3 x = t$

$$= \frac{1}{3}\int \frac{dt}{t^2} \Rightarrow -\frac{1}{3t} + c$$

$$= -\frac{1}{3\left(1 + \tan^2 x\right)} + c$$

67. C

Sol. $C_1 \to C_1 - bC_3, C_2 \to C_2 + aC_3$

$$\Delta = \begin{vmatrix} 1+a^2-b^2+2b^2 & 0 & -2b \\ 0 & 1-a^2+b^2+2a^2 & 2a \\ 2b-b+a^2b+b^3 & -2a+a-a^3-ab^2 & 1-a^2-b^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1+a^2+b^2 & 0 & -2b \\ 0 & 1+a^2 & 2a \\ b\left(b^2+a^2+1\right) & -a\left(1+a^2+b^2\right) & 1-a^2-b^2 \end{vmatrix}$$

$$= \left(1+a^2+b^2\right)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ b & -a & 1-a^2-b^2 \end{vmatrix}$$

$$= \left(1+a^2+b^2\right)^2 \left(\left(1-a^2-b^2\right)+2a^2-2b(-b)\right)$$

$$= \left(1+a^2+b^2\right)^3$$

Now, given that $ab = 1$

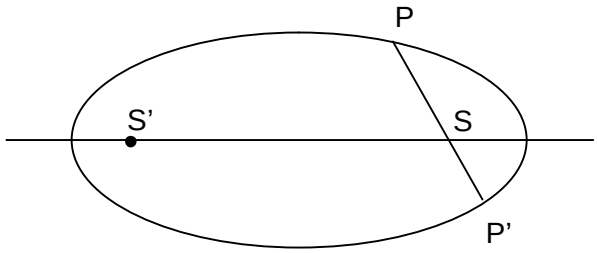
68. A

Sol. $e = \frac{3}{4}$

radius of auxillary circle

$= a = 5, b = \frac{5\sqrt{7}}{4}$

Now semi latus rectum
= HM (SP and SP')
Now SP' + S'P' = 2a
$S'P' = 6$



69. D

Sol. Let $M = \begin{bmatrix} a & p & x \\ b & q & y \\ c & r & z \end{bmatrix}$

$\Rightarrow t_r\left(M^T M\right) = a^2+b^2+c^2+p^2+q^2+r^2+x^2+y^2+z^2 = 5$

$\therefore$ Total number of ways $= {}^9P_2 + {}^9C_5 = 72+126 = 198,$ where

${}^9P_2 =$ Number of matrices when 2 comes once, 1 comes once and 0 comes 7 times

${}^9C_5 =$ Number of matrices when 1 comes 5 times and 0 comes 4 times.

70. D

Sol. Let

$$I = \int_1^2 xf(x)$$

$$\therefore I = \int_1^2 (3-x)f(3-x)dx = \int_1^2 (3-x)f(x)dx$$

$$\therefore 2I = \int_1^2 3f(x)dx$$

$$\Rightarrow I = \frac{3}{2}\int_1^2 f(x)dx = \frac{3K}{2}$$

71. A

Sol. We have $T_{r+1} = {}^{21}C_r \left(\sqrt[3]{\frac{a}{\sqrt{b}}}\right)^{21-r} \left(\sqrt{\frac{b}{\sqrt[3]{a}}}\right)^{r}$

$$= {}^{21}C_r (a)^{7-\frac{r}{2}} (b)^{\frac{2}{3}=-\frac{7}{2}}$$

Since the power of a and b are same,

$$\therefore 7 - \frac{r}{2} = \frac{2}{3}r - \frac{7}{2} \Rightarrow r = 9$$

72. C

Sol. 2nd blue result occur on or before 10th roll is equivalent to the occurrence of blue face at least twice in 10 rolls

$$n = 10,\ p = \frac{1}{3},\ q = \frac{2}{3}$$

$$p(m \geq 2) = 1 - (p(0) + p(1)) = \frac{3^9 - 2^{11}}{3^9}$$

73. B

Sol. Minimum distance lie along normal p(t)

on parabola $x^2 = 4y$

$\left(2t, t^2\right)$

Slope of tangent at p(t) = 1

equation of tangent to $x^2 = 4y$ at

$\left(2t, t^2\right)$

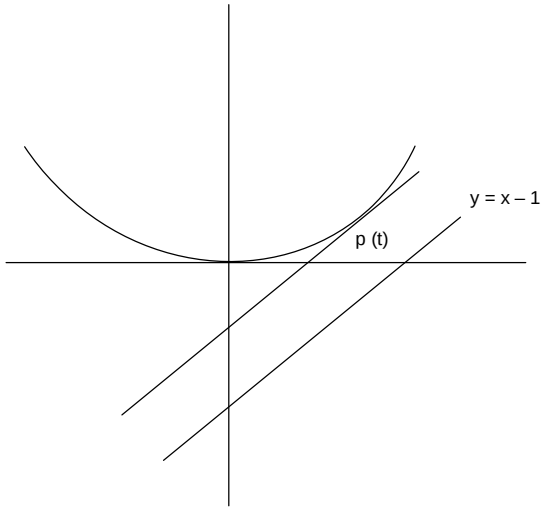
$x(2t) = 2.1\left(y + t^2\right)$

$tx = y + t^2$

slope = t = 1

p (2, 1)

sum of coordinate = 3



74. C

Sol. Let common difference be d.

$a_p = a_1 + (p-1)d, a_q = a_1 + (q-1)d,$

$a_r = a_1 + (r-1)d$

As a_p, a_q, a_r are in G.P.

$\therefore a_q = a_r = a_q - a_r$

$a_p - a_q = a_p - a_q$ (by law of proportions)

Or $\frac{a_q}{a_p} = \frac{a_p}{a_q} = \frac{a_1 + (q-1)d - a_1 - (r-1)d}{a_1 + (p-1)d - a_r - (q-1)d} = \frac{q-r}{p-q}$

Or $\frac{a_q}{a_p} - \frac{q-r}{p\ q} = \frac{r-q}{q\ \ p}$

75. C

Sol. In a 7 – digit number formed by using 1, 2 and 3, suppose that 1 appears x times, 2 appear y times and 3 appears z times. Then by hypothesis $x + 2y + 3z = 10$ and $x + y + z = 7$

Solving these equations we get $y + 2z = 3$ from which we get either

$y = 1, z = 1$ and $x = 5$ or $y = 3, z = 0$ and $x = 4$

Therefore, the total number is $\frac{7!}{5!} + \frac{7!}{4!3!} = 42 + 35 = 77$

76. B

Sol. The given equation is equivalent to $2\sin^2\left(\left(\frac{\pi}{2}\right)\cos^2 x\right) = 2\sin^2\left(\left(\frac{\pi}{2}\right)\sin 2x\right)$

$\Rightarrow \quad \cos^2 x = \sin 2x$

$\Rightarrow \quad \cos x(\cos x - 2\sin x) = 0$

$\Rightarrow \quad 1 - 2\tan x = 0$ as $\cos x \neq 0$

$\because \quad x \neq (2n+1)\frac{\pi}{2}$

$\Rightarrow \quad \tan x = \frac{1}{2}$

$\Rightarrow \quad \cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x} = \frac{3}{5}$

77. A

Sol. $x + iy = \frac{3(2 + \cos\theta - i\sin\theta)}{(2 + \cos\theta)^2 + \sin^2\theta}$

$= \frac{3(2 + \cos\theta) + i(-3\sin\theta)}{5 + 4\cos\theta}$

Comparing the real and imaginary parts we get

$$x^2+y^2=\frac{9(2+\cos\theta)^2+9\sin^2\theta}{(5+4\cos\theta)^2}$$

$$=\frac{9(5+4\cos\theta)}{(5+4\cos\theta)^2}=\frac{9}{5+4\cos\theta}$$

Also $4x-3=\frac{12(2+\cos\theta)}{5+4\cos\theta}-3=\frac{9}{5+4\cos\theta}$

Therefore $x^2+y^2=4x-3$

78. A

Sol. $\sin^{-1}\left(\sqrt{\frac{2-\sqrt{3}}{4}}\right)=\sin^{-1}\left(\sqrt{\frac{4-2\sqrt{3}}{8}}\right)=\sin^{-1}\left(\sqrt{\frac{\left(\sqrt{3}-1\right)^2}{8}}\right)=\sin^{-1}\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)=\frac{\pi}{12}$

$$\cos^{-1}\left(\frac{\sqrt{12}}{4}\right)=\cos^{-1}\left(\frac{2\sqrt{3}}{4}\right)=\frac{\pi}{6}$$

$$\sec^{-1}\left(\sqrt{2}\right)=\frac{\pi}{4}$$

$$\cos^{-1}\left[\cot\left\{\frac{\pi}{12}-\frac{\pi}{6}+\frac{\pi}{4}\right\}\right]=\cos^{-1}\left[\cot\left\{\frac{\pi}{2}\right\}\right]$$

$$=\cos^{-1}(O)=\frac{\pi}{2}$$

79. A

Sol. AB = A + B

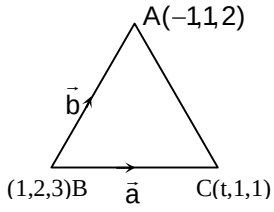
$\Rightarrow$ B = AB – A = A(B – I)

$\Rightarrow$ det(B) = det(A) det(B – I) = 0

$\Rightarrow$ det(B) = 0.

80. A

Sol. $A=\frac{1}{2}\left|\vec{a}\times\vec{b}\right|$ and $\left|\vec{a}\times\vec{b}\right|^2=\vec{a}^2\vec{b}^2-\left(\vec{a}\cdot\vec{b}\right)^2$



$\vec{a}=(t-1)\hat{i}-\hat{j}-2\hat{k}$; $\vec{b}=2\hat{i}+\hat{j}+\hat{k}$

$|\vec{a}|^2=(t-1)^2+1+4$; $|\vec{b}|^2=4+1+1=6$

$\vec{a}\cdot\vec{b}=2(t-1)-1-2=2t-5$

$|\vec{a}\times\vec{b}|^2=6[t^2-2t+6]-(4t^2+25-20t)$

$|\vec{a}\times\vec{b}|^2=2t^2+8t+11$ which is minimum at t = – 2

$|\vec{a}\times\vec{b}|^2_{min}=8-16+11=3$

$|\vec{a}\times\vec{b}|_{min}=\sqrt{3}$

SECTION – B

81. 00012.50

Sol. Observe that the x – coordinates of the intersection points of $y = x^3 + 8x^2 + 40x$ and $y = k$ are the 3 solutions of the Equation $-x^3 + 8x^2 + 40x = k$. Since these solutions are in geometric progression, let a, ar and ar^2 be the 3 distinct real roots of the cubic equation $x^3 - 8x^2 - 40x + k = 0$. By theory of equation:

$a + ar + ar^2 = 8,$ …(1)

$a^2r + a^2r^2 + a^2r^3 = -40$ …(2)

$-a^3r^3 = k$ …(3)

From (1) and (2), we have

$ar\left(a + ar + ar^2\right) = -40$

$ar \times 8 = -40$

$ar = -5$.

Substituting $ar = -5$ into (3) gives

$(-5)^3 = k$

$K = 125$

$\frac{k}{10} = 12.5$

82. 00004.00

Sol. Apply LMVT in [1, 2], $1 < e_1 < 2$

$f'(c) - \frac{f(2) - f_1}{2 \quad 1} > 2$

$f(2) \geq 4$ ….(1)

apply LMVT in $|2, 3|$, $2 < c_2 < 3$

$f'(c_2) = \frac{f(3) \quad f(2)}{3 - 2} \geq 2$

$7 - f(2) > 2 > f(2) < 5$ ....(2)

from (1) and (2) :

$4 \leq f(2) \leq 5$

83. 00003.00

Sol. $\cot 25^o \cot 35^o + \cos 85^o \cot 35^o - \cot 85^o \cot 25^o = \cot 25^o$

$\cot\left(60^o - 25^o\right) + \cot\left(60^o - 25^o\right)$

$\cot\left(60^o + 25^o\right) - \cot\left(60^o + 25^o\right)\cot 25^o = 3$

84. 00006.00

Sol. $x^2 + 9y^2 - 4x + 6y + 4 = 0$

$\Rightarrow \quad (x-2)^2 + 9\left(y^2 + \frac{2}{3}y + \frac{1}{9}\right) = 1$

$\Rightarrow \quad \frac{(x-2)^2}{1} + \frac{\left(y+\frac{1}{3}\right)^2}{\left(\frac{1}{9}\right)} = 1$

Its parametric equation is

$x = 2 + \cos\theta, y = -\frac{1}{3} + \frac{1}{3}\sin\theta$

Hence $4x - 9y = 8 + 4\cos\theta + 3 - 3\sin\theta$

$\Rightarrow$ minimum value of 4x-9y is 6

85. 00004.50

Sol. $R\cos A = \frac{3}{\sqrt{5}} \Rightarrow -R\cos B\cos C + R\sin B\sin C = \frac{3}{\sqrt{5}}$ and $2R\cos B\cos C = \frac{3}{\sqrt{5}}$

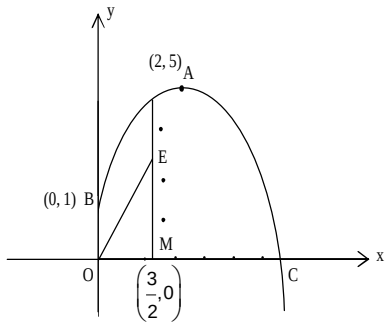
$\Rightarrow R\sin B\sin C = \frac{3}{\sqrt{5}} + \frac{3}{2\sqrt{5}} = \frac{9}{2\sqrt{5}}$

$\sin B\sin C = \frac{9}{2\sqrt{5}} \times \sqrt{\frac{5}{61}} = \frac{9}{2\sqrt{61}}$.

86. 00008.00

Sol. $y = 1 + 4x - x^2 = 5 - (x-2)^2$

$\Rightarrow (x-2)^2 = 5 - y = -(y-5)$. This is parabola with vertex A at (2, 5) and facing downwards.



Area OMEAB $= \int_0^{\frac{3}{2}} \left(1 + 4x - x^2\right) dx = \left[x + 2x^2 - \frac{x^3}{3}\right]_0^{\frac{3}{2}}$

$= \frac{3}{2} + \frac{9}{2} - \frac{27}{24} = \frac{39}{8}$ OE is line $y = mx$, so that E is $\left(\frac{3}{2}, \frac{3m}{2}\right)$

Area of Δ OME $= \frac{1}{2} \times \frac{3}{2} \times \frac{3m}{2} = \frac{1}{2} \times \frac{39}{8}$ (By problem)

$\Rightarrow \quad m = \frac{13}{6}$.

87. 00019.00

Sol. $\left.\frac{dy}{dx}\right|_{(\alpha,\beta)} = \frac{-\alpha^2 - 3}{\left(\alpha^2 - 3\right)^2} f$

Given that :

$$\frac{-\alpha^2-3}{\left(\alpha^2-3\right)^2}=-\frac{1}{3}$$

$\Rightarrow \alpha = 0, \pm 3 \qquad (\alpha \neq 0)$

$\Rightarrow \beta = \pm\frac{1}{2}. \qquad (\beta \neq 0)$

$|6\alpha + 2\beta| = 19$

88. 00003.50

Sol. $x^2+(y-1)^2 = x^2+(y+2)^2$

$-2y+1 = 4y+4$

$6y = -3 \Rightarrow y = -\frac{1}{2}$

$x^2+y^2 = \frac{25}{4} \Rightarrow x^2 = \frac{24}{4} = 6 \Rightarrow \quad z = \pm\sqrt{6}-\frac{i}{2}$

$|z+3i| = \sqrt{6+\frac{25}{4}} = \sqrt{\frac{49}{4}}$

$|z+3i| = \frac{7}{2}$

89. 00019.00

Sol. $k_1p+7 = 330$

$k_2p+11 = 486$

$k_3p+15 = 604$

$k_1p = 323 = (19)(17)$

$k_2p = 475 = (19)(25)$

$k_3p = 589 = (19)(31)$

Hence p is 19

90. 00005.00

Sol.
$$\lim_{n\to\infty}\frac{\int_0^{\frac{1}{n}} x^{x+k-1}}{\left(\frac{1}{n}\right)^k}dx = \lim_{t\to 0}\frac{\int_0^{t} x^{x+k-1}dx}{t^k}$$

Taking t as continuous variable (as $t \to 0$)

$$\lim_{n\to\infty}\frac{t^{t+k-1}}{k\,t^{k-1}} = \lim_{t\to 0}\frac{1}{k}t^t = \frac{1}{k}.$$